



Smart Traffic Light Controller with Adaptive Flow

Course: Data Structures — Fall 2025

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Project #: 8

The Problem: Gridlock in Our Cities

Our urban centers are facing an escalating challenge: severe traffic congestion during peak hours. The current infrastructure, largely relying on fixed-timer traffic lights, is simply inadequate for dynamic urban environments. This leads to a cascade of negative effects:

Long Queues & Delays

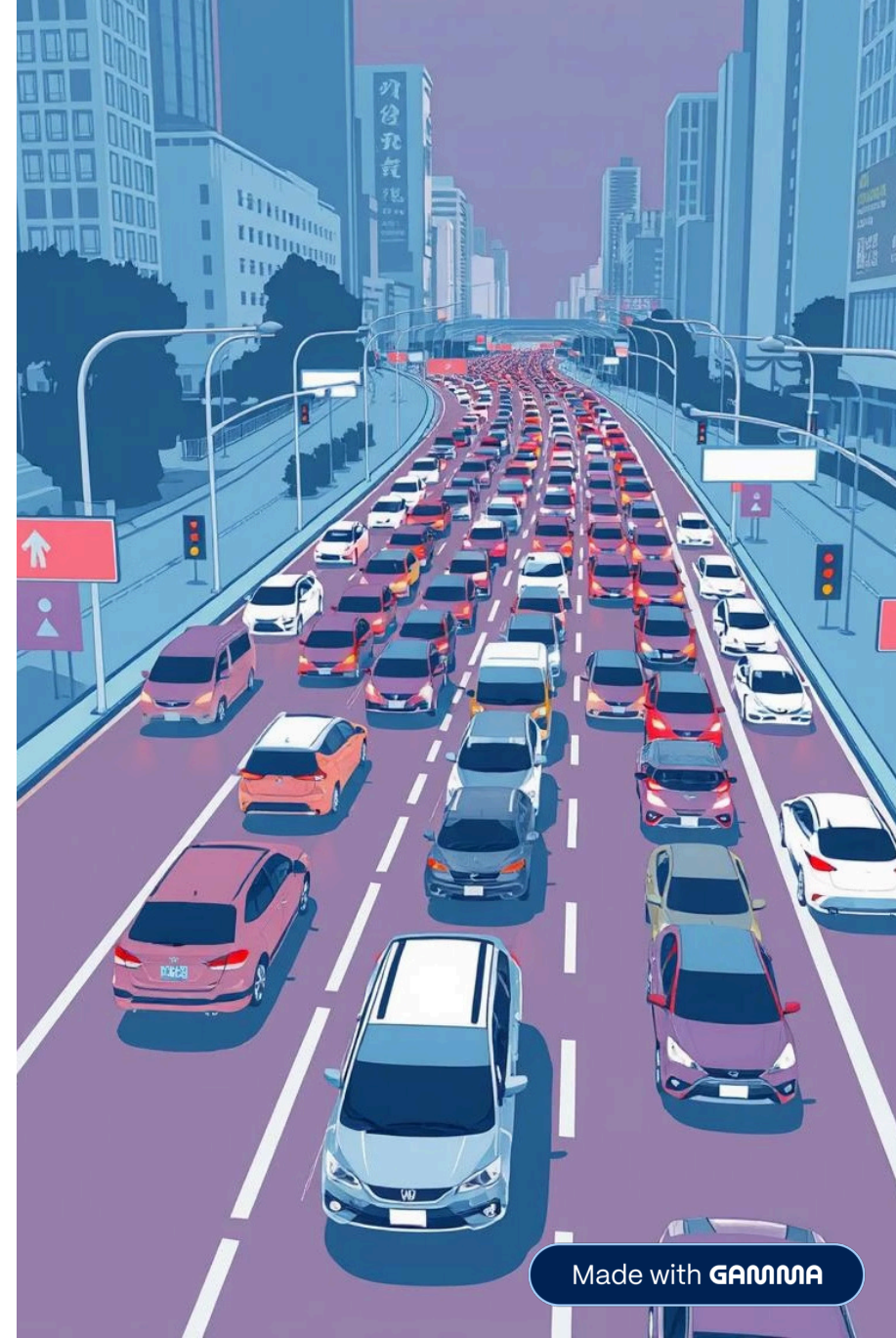
Vehicles spend excessive time waiting, leading to frustration and lost productivity.

Inefficient Green-Time Allocation

Fixed timers fail to adapt to real-time traffic demands, resulting in wasted green light phases for empty lanes.

Increased Waiting Times

Drivers experience unpredictable and often prolonged delays, impacting daily commutes and logistics.



Our Proposed Solution: Dynamic Traffic Management

We propose a C++ simulation designed to revolutionize urban traffic control. This system moves beyond static timers, introducing a dynamic, adaptive approach that responds to real-time traffic conditions.

→ Real-time Queue Monitoring

The system continuously reads and analyzes vehicle queue lengths at each lane approaching an intersection.

→ Dynamic Lane Selection

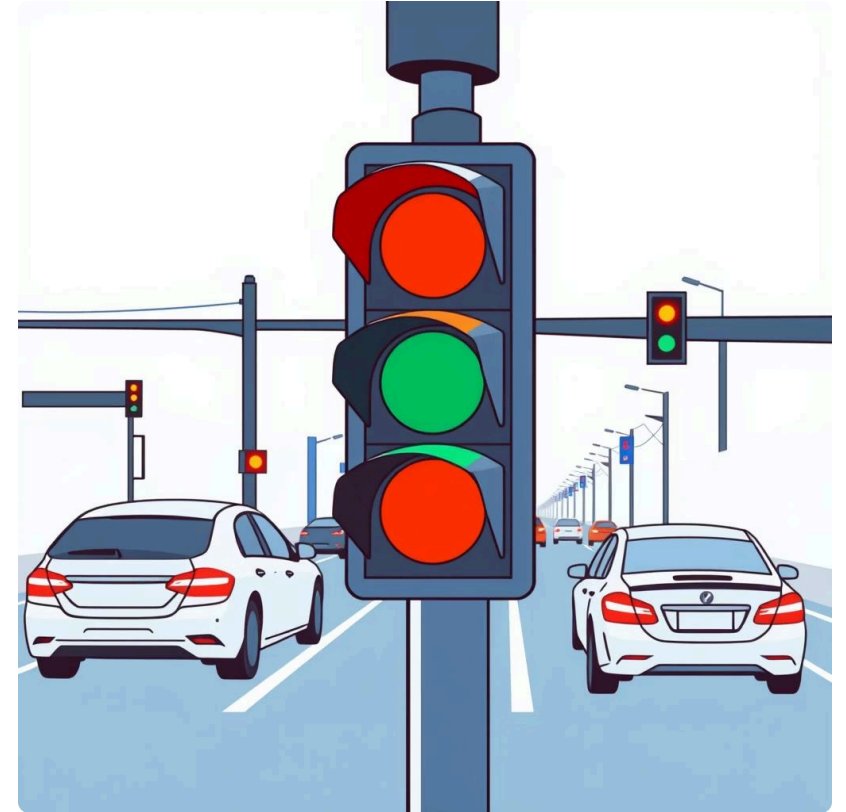
Based on queue data, the controller intelligently selects the lane requiring a green light, prioritizing congestion.

→ Adaptive Green-Light Duration

Green light durations are adjusted in real-time, optimizing flow and minimizing unnecessary waits.

→ Reduced Average Waiting Time

The ultimate goal: significantly cut down on vehicle waiting times across the simulated network.



Core Functional Requirements

To achieve an effective and realistic simulation, our system incorporates several critical functional requirements, each contributing to a comprehensive traffic management solution.

Vehicle Arrival & Lane Queues

Accurate simulation of vehicle generation and their orderly queuing at intersection approaches.

Adaptive Green-Light Logic

The intelligence behind dynamic signal timing based on real-time data.

Intersectional Graph Representation

Modeling the city's road network as a graph, allowing complex routing.

Vehicle Movement Between Intersections

Simulating the flow of cars from one intersection to the next.

Performance Metrics Calculation

Measuring key indicators like average waiting time, queue lengths, and throughput.

Console-Based Simulation

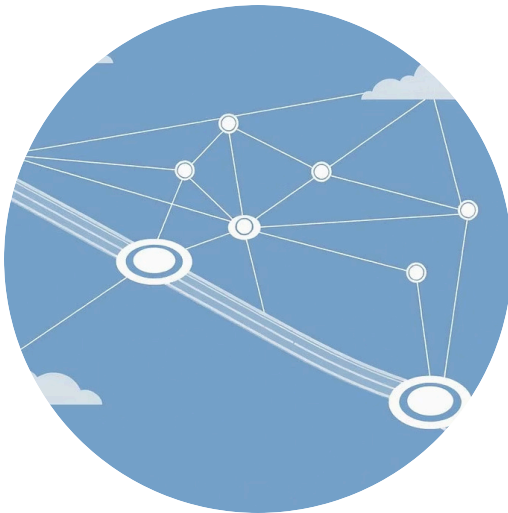
Providing a clear, real-time textual output of the simulation's state and performance.

Essential Data Structures for Traffic Management

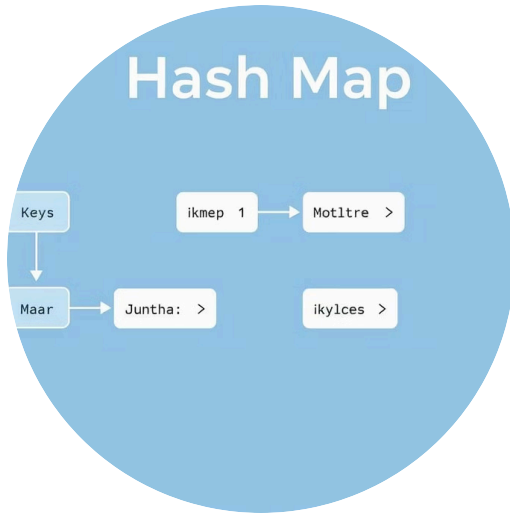
The efficiency and accuracy of our traffic simulation heavily rely on the strategic selection and implementation of appropriate data structures. Each structure serves a distinct purpose, optimizing specific aspects of the system.



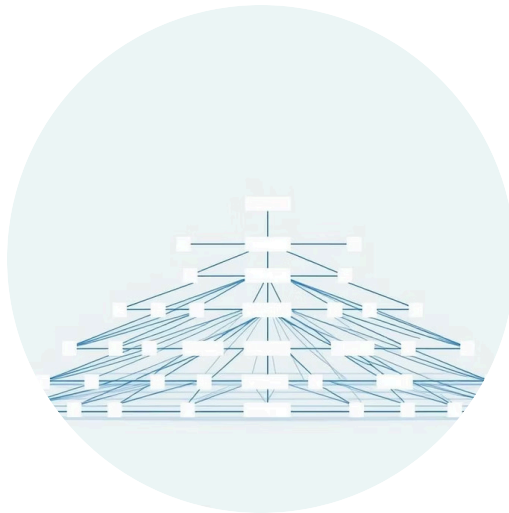
Queue (for Lanes)
Models vehicle waiting lines at each traffic lane.



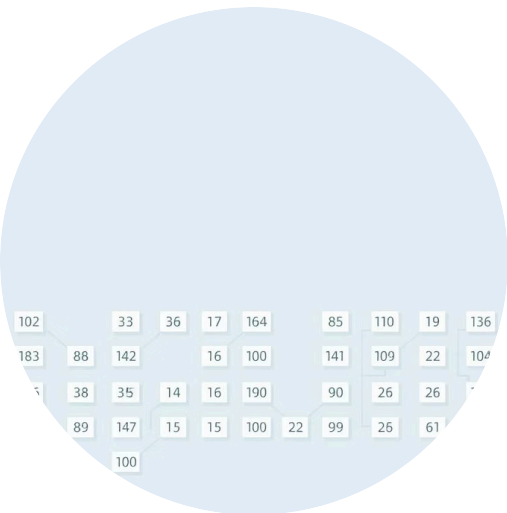
Graph (for Intersections)
Represents the interconnected network of city intersections.



HashMap (for Statistics)
Stores and retrieves performance statistics for each intersection.



Max-Heap (for Lane Selection)
Efficiently identifies the lane with the highest congestion for priority.



Arrays (for Cycle Logs)
Used for simple and fast logging of simulation cycles and events.

Justification: The 'Why' Behind Our Data Structure Choices

Every data structure in our simulation was chosen for its specific properties that align with real-world traffic dynamics and computational efficiency.



Queue: Real-life FIFO Flow

Vehicles at a traffic light naturally follow a First-In, First-Out (FIFO) principle. A queue perfectly models this behavior, ensuring fairness and accurate simulation of waiting lines.



Graph: City Network Representation

A city's road system, with its intersections and connecting roads, is inherently a graph. This structure allows us to represent complex routing, one-way streets, and varying travel times between intersections.



HashMap: Fast Per-Signal Stat Lookup

For quickly accessing and updating statistics (like average wait times or total vehicles passed) for each individual intersection, a HashMap provides average $O(1)$ time complexity, crucial for real-time performance.



Priority Queue (Max-Heap): Fast Lane Selection

To implement adaptive logic, we need to efficiently select the lane with the highest congestion. A max-heap allows us to extract the maximum element (longest queue) in $O(\log n)$ time, ensuring quick decision-making.

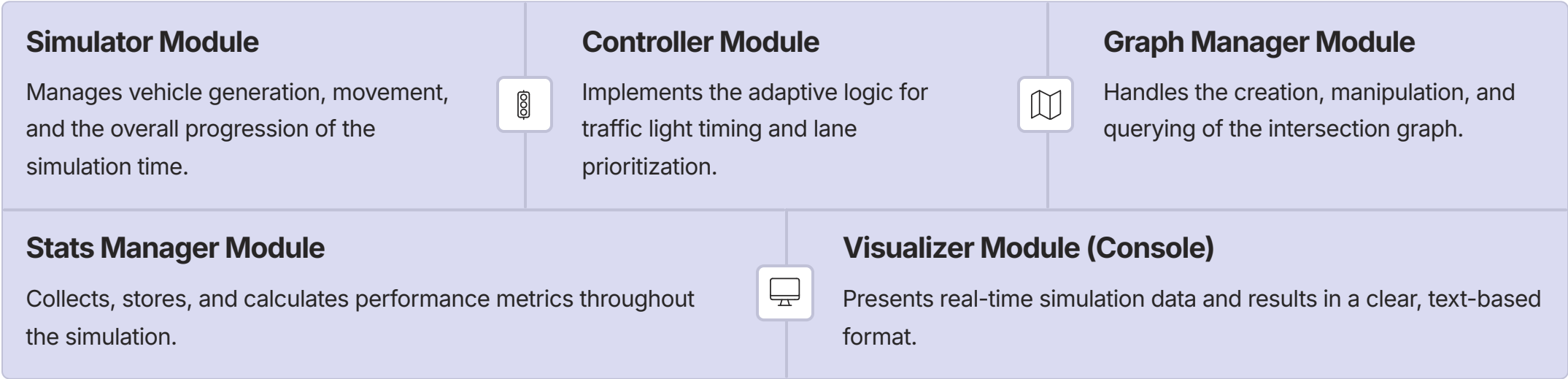


Arrays: Simple and Fast for Logging

For storing sequential data, such as a log of cycle metrics or events, arrays offer direct access and minimal overhead, making them ideal for simple data accumulation.

System Architecture: Modular Design for Clarity and Scalability

Our Smart Traffic Light Controller is built on a modular architecture, breaking down complex functionalities into distinct, manageable components. This approach enhances maintainability, facilitates parallel development, and allows for easier future expansion.



Adaptive Algorithm Overview: The Flow of Intelligence

The heart of our system is an adaptive algorithm that continuously monitors, analyzes, and adjusts traffic flow. This step-by-step process ensures optimal green-light allocation and minimizes congestion.

01

Generate Vehicle Arrivals

New vehicles are introduced into the simulation probabilistically, reflecting real-world traffic patterns.

02

Read Queue Lengths

The system reads the current queue lengths for all lanes approaching an intersection.

03

Insert Lane Priorities into Max-Heap

Queue lengths are used to determine lane priorities, which are then inserted into a max-heap.

04

Select Lane with Highest Load

The max-heap is queried to efficiently identify the lane with the greatest number of waiting vehicles.

05

Compute Green-Time

An adaptive formula calculates the optimal green-light duration for the selected lane, proportional to its load.

06

Dequeue Vehicles

Vehicles are processed and removed from the queue according to the computed green-time.

07

Update Stats in HashMap

Real-time statistics for the intersection are updated, including throughput and waiting times.

08

Route Vehicles Across Graph Edges

Departing vehicles are routed to their next destination intersection based on the graph structure.

09

Store Metrics in Arrays

Key performance indicators are logged for historical analysis and overall simulation evaluation.

10

Repeat Next Cycle

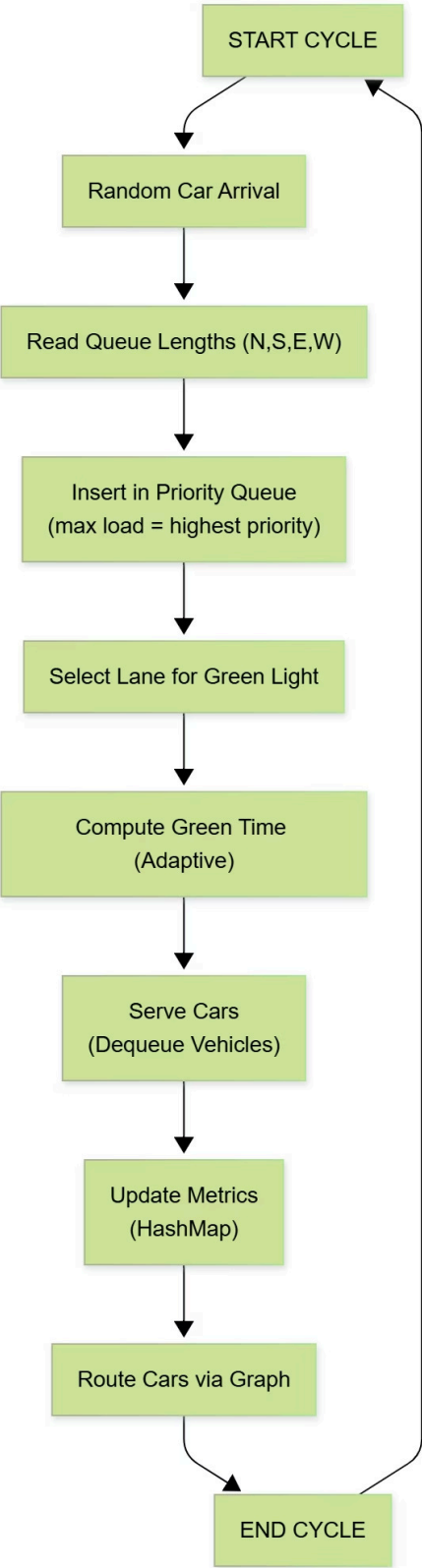
The entire process reiterates, ensuring continuous adaptation to evolving traffic conditions.

Adaptive Cycle in Action

This diagram illustrates a single cycle of our adaptive traffic control algorithm, demonstrating how real-time data drives dynamic decisions.

Example Scenario:

- **North:** 10 cars → Longer green time allocated.
- **East:** 0 cars → Lane is skipped, preventing wasted green time.
- **West:** Medium load → Medium green time allocated.
- Statistics are updated, and vehicles move to connected intersections via the graph, initiating the next cycle.



Conclusion: Towards Smarter, More Efficient Cities

Our Smart Traffic Light Controller simulation represents a significant step forward in urban traffic management. By leveraging the power of data structures and adaptive algorithms, we demonstrate a robust solution for alleviating congestion.

Enhanced Traffic Efficiency

The dynamic allocation of green light phases significantly improves overall traffic flow.

Reduced Waiting Times

By prioritizing congested lanes, the system drastically cuts down on vehicle wait times.

Real-World Applicability

Our simulation provides a foundation for developing intelligent traffic systems that can adapt to changing urban demands.



This project showcases the potential of computer science principles to address complex real-world problems, paving the way for more responsive and sustainable urban environments.